# 2025 Integration Bee

Paly Math Club

May 2025

## 1 Qualifiers (10Q) easy – hard

1. $\int_3^5 x dx$
2. $\int (\sin(x) + \cos(x)) dx$
3. $\int (\sec^2(x) - \tan^2(x)) dx$
4. $\int (\frac{1 + \cos(x)}{x + \sin(x)}) dx$
5. $\int_0^1 (5x^3 - 2x + 7)\, dx$
6. $\int_1^4 x^{-\frac{1}{2}}\, dx$
7. $\int_{-5}^5 (\frac{x^{27} + x^9}{\sqrt{x^2 \ln(x^2) + \cos(\cos(x^2))}}) dx$
8. $\int_0^1 x^{e^2}\, dx$
9. $\int_0^e \ln x\, dx$
10. $\int_0^\infty \frac{1}{e^x + 1} dx$

## 2 Round of 16, best of 1 (easy - 8+4Q)

1. $\int_0^1 (\sqrt{4-x^2})dx$
2. $\int \frac{e^x}{1+e^{2x}}\,dx$
3. $\int \sin^3 x\,dx$
4. $\int \arctan x\,dx$
5. $\int_{-1}^{1} (1+x)(1+x^2)\,dx$
6. $\int_1^4 \frac{6}{\sqrt{x}\,(1+\sqrt{x})^3}\,dx$
7. $\int_1^2 e^{\,3\ln(x^2)-2\ln(x)}\,dx$
8. $\int_2^3 \frac{x+5}{(4-x)^4}\,dx$
9. $\int_0^{2\pi} \sin x \cos x \tan x \sec x \csc x \cot x\,dx$
10. $\int_0^1 \Big(\sum_{k=1}^{\infty} (-1)^k\, x^{2k}\Big)\,dx$

## 3 Quarterfinals, best of 1 (medium - 4+2Q)

1. $\int x^2 \ln x\,dx$
2. $\int xe^{3x}\,dx$
3. $\int_1^{e^e} \frac{\log\left(x^{\log(x^x)}\right)}{x^2}\,dx$
4. $\int \frac{dx}{1+\sin x}$

5. $\int_0^{\pi} (x + \sin(x) + x\cos(x) + \sin(x)\cos(x))dx$

6. $\int_0^2 (\sqrt{x}\min(\sqrt{x}, \sqrt{2-x}))dx$

## 4 Semifinals, best of 3 (medium - 6Q)

1. $\int (\frac{x^8}{\sqrt{x^{18}-1}})dx$

2. $\int (\sin^3(x) + \cos^3(x))dx$

3. $\int \frac{3x^3 - 6x^2 + 7x - 9}{(x-2)^2(x^2+1)}dx$

4. $\int_0^1 (\sqrt{1-\sqrt{x}})dx$

5. $\int \frac{\tan x + \cot x}{\sec x \csc x}\,dx$

6. $\int_0^{40} \{x\}\lfloor x \rfloor\,dx$

7. $\int \sqrt[\frac{3}{1}]{x\sqrt[\frac{4}{2}]{x\sqrt[\frac{5}{3}]{x\sqrt[\frac{6}{4}]{\cdots}}}}\quad dx$

## 5 Finals, best of 5 (hard - 5Q)

1. $\int_4^6 (\frac{d}{dx}[\frac{(x-4)^2}{2\cdot 3} + \frac{(x-4)^3}{3\cdot 3^2} + \frac{(x-4)^4}{4\cdot 3^3} + \cdots + \frac{(x-4)^n}{n\cdot 3^{n+1}}])dx$

2. $\int \frac{dx}{\sin x + \cos x}$

3. $\int x\,e^x \sin x\,dx$

4. $\int_0^1 \frac{\ln(1+x)}{x}\,dx$ (hint if no one can solve: infinite series)

5. $\int_1^2 (x^{x^x+x}\ln^2(x) + x^{x^x+x}\ln(x) + x^{x^x+x-1})dx$

6. $\int \sin x \, \sinh x \, dx$

## 6 Tiebreakers (3Q)

1. $\int_1^\infty x^5 e^{-x}\, dx$

2. $\int \sqrt{\dfrac{x}{4-x}}\, dx$

3. $\int_0^\pi \dfrac{1+\cos x}{1+\sin x}\, dx$

4. $\int_0^{\pi/2} \sin^n x\, dx$ (where $n \geq 3$ is an odd number)